

KING FAHAD CENTRAL HOSPITAL / JAZAN HEALTH CLUSTER

Comprehensive / Primary Stroke Center Program

EARLY REHABILITATION ASSESSMENT AND REFERRAL PROTOCOL FOR STROKE PATIENTS

Clinical Practice Protocol

Prepared in Alignment with American Heart Association / American Stroke Association
Guidelines and Get With The Guidelines—Stroke / Target: StrokeSM Best Practice Standards

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1. Purpose

This protocol establishes a standardized, evidence-based process for the early screening, assessment, and referral of patients to rehabilitation services following acute ischemic or hemorrhagic stroke. It is designed to ensure timely identification of rehabilitation needs, minimize complications associated with immobility, promote functional recovery, and support compliance with American Heart Association/American Stroke Association (AHA/ASA) guidelines and Get With The Guidelines–Stroke (GWTG-Stroke) performance measures required for Primary Stroke Center (PSC), Thrombectomy-Capable Stroke Center (TSC), and Comprehensive Stroke Center (CSC) accreditation.

This protocol operationalizes recommendations from the 2016 AHA/ASA Guidelines for Adult Stroke Rehabilitation and Recovery and the 2019 AHA/ASA Guidelines for the Early Management of Patients with Acute Ischemic Stroke, and is intended for use by the multidisciplinary stroke care team across the continuum of inpatient care.

2. Scope

This protocol applies to all adult patients (age ≥ 18 years) admitted with a confirmed or suspected diagnosis of acute ischemic stroke, transient ischemic attack (TIA), intracerebral hemorrhage (ICH), or subarachnoid hemorrhage (SAH), across the Emergency Department, Neurological/Stroke Intensive Care Unit, and inpatient stroke or medical-surgical units. It applies to nursing, physician, advanced practice provider, physical therapy (PT), occupational therapy (OT), speech-language pathology (SLP), case management, and rehabilitation medicine staff.

3. Policy Statement

It is the policy of King Fahad Central Hospital / Jazan Health Cluster that every patient admitted with acute stroke receives a structured rehabilitation needs screen within 24–48 hours of hospital admission (or as soon as medically stable), formal interdisciplinary rehabilitation assessment when indicated, and referral to the most appropriate level of post-acute rehabilitation care prior to discharge, consistent with AHA/ASA Class I recommendations supporting early, safe mobilization and structured rehabilitation planning.

Early mobilization refers to structured out-of-bed activity typically initiated within 24–48 hours of stroke onset, distinct from very early (within 24 hours) high-dose mobilization, which is not recommended per the AVERT trial findings incorporated into current AHA/ASA guidance.

4. Definitions

Term	Definition
Early Mobilization	Progressive, structured out-of-bed activity (sitting, standing, ambulating) initiated after medical stabilization, generally within 24–48 hours of stroke onset, and titrated to patient tolerance.

Term	Definition
Rehabilitation Screen	A brief, standardized bedside evaluation performed by nursing or therapy staff to identify functional, swallowing, cognitive, and communication deficits requiring further formal assessment.
Formal Rehabilitation Assessment	A comprehensive evaluation performed by licensed PT, OT, and/or SLP clinicians to establish baseline function, rehabilitation potential, and discharge disposition recommendations.
NIHSS	National Institutes of Health Stroke Scale – standardized tool used to quantify stroke severity and neurological deficit.
Modified Rankin Scale (mRS)	A 7-point scale (0–6) measuring degree of disability or dependence in activities of daily living post-stroke.
Inpatient Rehabilitation Facility (IRF)	A post-acute care setting providing intensive, multidisciplinary rehabilitation (typically ≥ 3 hours/day, 5 days/week) for patients able to tolerate and benefit from intensive therapy.
Skilled Nursing Facility (SNF)	A post-acute setting providing skilled nursing and lower-intensity rehabilitation therapy for patients not yet able to tolerate IRF-level intensity.
LTACH	Long-Term Acute Care Hospital – setting for medically complex patients requiring extended acute-level care with concurrent rehabilitation.

5. Clinical Rationale and Evidence Basis

Early rehabilitation assessment reduces stroke-related complications including venous thromboembolism, aspiration pneumonia, pressure injury, contracture, and deconditioning, while supporting neuroplasticity-driven recovery. AHA/ASA guidelines recommend that all stroke patients be evaluated for rehabilitation needs and that safe, early mobilization be initiated as part of routine post-stroke care.

- Structured rehabilitation assessment supports accurate triage to the appropriate level of post-acute care, reducing unnecessary length of stay and readmission risk.
- Dysphagia screening prior to any oral intake reduces aspiration pneumonia risk and is a required GWTG-Stroke performance measure.
- Early, progressive mobilization (avoiding high-dose activity within the first 24 hours) is associated with improved functional outcomes without increased adverse events when appropriately screened.
- Interdisciplinary rehabilitation planning initiated early in the inpatient stay is associated with shorter time to appropriate post-acute placement and improved patient/family engagement in discharge planning.

6. Assessment and Referral Timeline

The following timeline reflects minimum expected practice standards. Individual patient status (hemodynamic instability, elevated intracranial pressure, active hemorrhage extension, mechanical ventilation) may require deferral of specific steps; deferrals must be documented with clinical rationale and reassessed daily.

Timeframe	Required Action	Responsible Discipline
Within 4 hours of admission / arrival to unit	NIHSS documented; dysphagia (swallow) screen completed prior to any oral intake, medication, or food/fluids	RN / Physician / APP
Within 24 hours (once medically stable)	Bedside functional mobility and fall-risk screen; assess readiness for out-of-bed activity	RN, with PT consult as indicated
Within 24–48 hours	PT, OT, and SLP consults placed for all patients with NIHSS ≥ 1 or documented functional/communication/swallowing deficit	Attending Physician / Stroke APP
Within 48 hours of consult order	Formal PT/OT/SLP evaluations completed; baseline functional status and rehabilitation potential documented	PT / OT / SLP
Within 72 hours of admission	Interdisciplinary rehabilitation goals established; preliminary discharge disposition recommendation documented	Interdisciplinary Team / Case Management
48–72 hours prior to anticipated discharge	Rehabilitation referral submitted to appropriate post-acute level of care; family/caregiver education initiated	Case Management / Social Work
Prior to discharge	Final disposition confirmed; rehabilitation records, mRS, and functional status transmitted to receiving facility	Case Management / Attending Physician

7. Standardized Screening and Assessment Tools

7.1 Dysphagia Screening

All stroke patients must undergo a validated bedside dysphagia screen prior to any oral intake of food, fluid, or medication, regardless of stroke severity. A validated tool (e.g., a standardized water swallow protocol) must be used; nil per os (NPO) status is maintained until the screen is passed. Failed screens require formal SLP swallow evaluation, with consideration of instrumental assessment (videofluoroscopic swallow study or fiberoptic endoscopic evaluation of swallowing) as clinically indicated.

7.2 Functional Mobility and Fall Risk

Nursing staff complete a standardized fall-risk and functional mobility screen within 24 hours of admission. Any impairment in bed mobility, transfers, sitting balance, or ambulation triggers a PT consult.

7.3 Cognitive and Communication Screening

Patients are screened for aphasia, dysarthria, apraxia, neglect, and cognitive impairment using standardized bedside tools. Any positive finding triggers formal SLP and/or OT evaluation, including assessment of cognitive-communication function and safety awareness.

7.4 Standardized Outcome Measures

- National Institutes of Health Stroke Scale (NIHSS) – documented on admission, at defined intervals, and with any neurological change.

- Modified Rankin Scale (mRS) – documented on admission and at discharge to support disposition planning and outcome tracking.
- Functional Independence Measure (FIM) or equivalent validated functional scale – documented by PT/OT at initial evaluation and prior to discharge.
- Barthel Index (optional/supplemental) – may be used to further characterize activities-of-daily-living independence.

8. Rehabilitation Referral Criteria and Disposition Pathways

Disposition recommendations are made collaboratively by the interdisciplinary team based on functional assessment findings, medical stability, cognitive status, family/caregiver support, and patient preference. The following criteria guide, but do not replace, individualized clinical judgment.

Disposition	General Referral Criteria
Inpatient Rehabilitation Facility (IRF)	Medically stable; able to actively participate in and tolerate ≥ 3 hours/day of therapy across at least two disciplines; realistic potential for functional gain; requires 24-hour physician and nursing oversight.
Skilled Nursing Facility (SNF) with Rehabilitation	Medically stable; unable to tolerate IRF-level therapy intensity; requires skilled nursing plus lower-intensity, ongoing rehabilitation therapy.
Long-Term Acute Care Hospital (LTACH)	Medically complex (e.g., ventilator-dependent, requiring ongoing intensive medical management) with concurrent rehabilitation needs anticipated to extend beyond standard acute stay.
Home with Outpatient Therapy	Mild deficits; medically stable; adequate home support and safety; able to access outpatient PT/OT/SLP services.
Home Health Services	Homebound status; moderate deficits requiring skilled therapy or nursing services delivered in the home setting.
Hospice / Palliative Care Referral	Consistent with patient/family goals of care in the setting of severe, non-recoverable deficit or comorbid terminal condition; addressed via palliative care consultation.

All referral decisions and the clinical rationale supporting the selected level of care must be documented in the medical record, including consideration of patient/family preference and psychosocial factors identified by Social Work/Case Management.

9. Early Mobilization Safety Parameters

Prior to initiating out-of-bed activity, the following parameters must be reviewed and documented as stable:

- Systolic blood pressure within physician-ordered parameters (typically $< 220/120$ mmHg unless post-thrombolysis/thrombectomy parameters apply, or as individually ordered).
- Oxygen saturation $\geq 92\%$ on prescribed supplemental oxygen, if applicable.
- Heart rate and cardiac rhythm stable without new arrhythmia.
- No evidence of clinical deterioration, worsening NIHSS, or suspected hemorrhagic transformation.

- Absence of contraindications such as unstable fracture, active deep vein thrombosis without anticoagulation, or physician hold order.

Mobilization is deferred, and the deferral documented with rationale and reassessment plan, if any of the above parameters are not met. Very early, high-frequency, high-dose mobilization within the first 24 hours is not recommended and should be avoided in favor of structured, progressive activity beginning at 24–48 hours.

10. Roles and Responsibilities

Role	Key Responsibilities
Attending Physician / APP	Confirms diagnosis and stability; orders PT/OT/SLP consults; sets mobilization parameters; reviews and approves final disposition.
Bedside Registered Nurse	Performs dysphagia and fall-risk screens; documents NIHSS trends; initiates safe mobilization within ordered parameters; escalates status changes.
Physical Therapist	Evaluates mobility, balance, and gait; establishes mobilization plan; recommends level of rehabilitation care based on functional findings.
Occupational Therapist	Evaluates activities of daily living, upper extremity function, cognition, and neglect; recommends adaptive equipment and safety strategies.
Speech-Language Pathologist	Performs formal swallow, communication, and cognitive-linguistic evaluation; recommends diet modification and communication strategies.
Case Manager / Social Worker	Coordinates interdisciplinary disposition planning; facilitates referral to receiving facility; addresses psychosocial and insurance authorization needs; leads family education.
Rehabilitation Medicine / Physiatrist (if available)	Provides consultative guidance on complex rehabilitation needs and appropriate level-of-care determination.

11. Documentation Requirements

Complete and timely documentation is required to support clinical continuity, quality reporting, and accreditation review. At minimum, the medical record must include:

1. Time and result of initial dysphagia screen, prior to any oral intake.
2. Admission and serial NIHSS scores.
3. Date/time of PT, OT, and SLP consult orders and completed evaluations.
4. Documented rationale for any delay or deferral of screening, mobilization, or consult beyond protocol timeframes.
5. Interdisciplinary rehabilitation goals and functional outcome measures (e.g., FIM, mRS).
6. Final discharge disposition with supporting clinical rationale.
7. Evidence of patient/family education and participation in discharge planning.

12. Quality Metrics and Performance Monitoring

The Stroke Program Committee monitors compliance with this protocol through routine chart audit and integration with Get With The Guidelines–Stroke data submission. Core metrics tracked include:

- Percentage of patients receiving a dysphagia screen prior to any oral intake.
- Percentage of patients with rehabilitation assessment (PT/OT/SLP consult) completed within 24–48 hours of admission.
- Median time from admission to first out-of-bed mobilization.
- Percentage of patients with documented discharge disposition and functional outcome measures (mRS at discharge).
- Rate of hospital-acquired complications associated with immobility (aspiration pneumonia, venous thromboembolism, pressure injury) among the stroke population.
- Readmission rate within 30 days stratified by discharge disposition.

Results are reported quarterly to the Stroke Program Committee and Medical Executive Committee, with corrective action plans initiated for any metric falling below the target compliance threshold established by the program (recommended minimum 90% compliance for time-sensitive measures).

13. Related Policies and References

Related Institutional Policies

- Acute Stroke Code / Thrombolytic and Thrombectomy Administration Protocol
- Dysphagia Screening and Diet Modification Policy
- Fall Prevention and Safe Patient Mobilization Policy
- Discharge Planning and Post-Acute Care Referral Policy

Clinical Guideline References

- Winstein CJ, et al. Guidelines for Adult Stroke Rehabilitation and Recovery: A Guideline for Healthcare Professionals from the American Heart Association/American Stroke Association. *Stroke*. 2016;47(6):e98–e169.
- Powers WJ, et al. Guidelines for the Early Management of Patients with Acute Ischemic Stroke. *Stroke*. 2019;50(12):e344–e418.
- AVERT Trial Collaboration Group. Efficacy and safety of very early mobilisation within 24 h of stroke onset (AVERT): a randomised controlled trial. *Lancet*. 2015;386(9988):46–55.
- American Heart Association / American Stroke Association. Get With The Guidelines–Stroke Program Manual and Target: StrokeSM Best Practice Strategies.

This protocol is intended to support, and should be used in conjunction with, current AHA/ASA guidelines and institutional stroke center accreditation requirements. Clinicians should exercise independent clinical judgment for individual patient circumstances.

14. Review and Approval

Role	Name / Title	Signature	Date
Stroke Program Medical Director			
Director of Rehabilitation Services			
Chief Nursing Officer / Nursing Director			
Stroke Program Coordinator			
Chair, Stroke Program Committee			